

Total No. of Questions : 8]

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Roll No

EE-7004(3)/EX-7004(4)-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

**Renewable and Non-Conventional Energy
Systems**

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) Assume any missing data by putting a note at appropriate place.
iii) Draw neat sketches wherever necessary.
1. a) Briefly discuss about various non-conventional energy sources, different methods of extracting energy from them and the relative cost considerations and economics. 7
b) Explain the basic working of Fuel Cells and mention the types. 4
c) Define the following: 3
i) Energy conservation
ii) Energy efficiency
 2. a) Describe in brief the main considerations in selecting a site for wind energy forms. 7
b) Explain about yaw control and pitch control in a horizontal wind generator with neat sketches. 4
c) Explain the principle of Wind Energy Conversion. 3

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3. a) Discuss the issues related to grid interconnection of wind farms. 7
b) Calculate the maximum power output of a 15-m diameter wind turbine at one atmospheric pressure and wind speed of 12 m/s. 4
c) Explain the small hydro system. What is the importance of this system in the present day energy scenario? 3
4. a) Define solar irradiance, solar constant, extra-terrestrial and terrestrial radiations. 7
b) Explain the working of a Grid interactive solar pv system with the help of a neat sketch. 4
c) Explain the electrical characteristics of a solar cell with the help of a neat labelled sketch. Also explain how maximum efficiency of a solar cell is achieved. 3
5. a) Explain the following two forms of biomass conversion technologies: 7
i) Anaerobic digestion
ii) Ethanol Fermentation
b) List the eight major factors which affect the quality and quantity of biogas. 4
c) Compare relative performances of floating drum and fixed-dome-type biogas plants. 3
6. a) Explain briefly the various types of geothermal resources available for power generation. 7

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- b) Mention the reasons for your selection of the following power plants for base load or peak load operation. 4

Type of power plant	Reason for using for Base load operation	Reason for using for Peak load operation
Geothermal		
OTEC		
Nuclear		
Wind		

- c) Explain the basic principle of ocean thermal energy. 3
7. a) What design characteristics are important while selecting a motor to conserve energy? 7
- b) Explain the difference between band gap and barrier voltage in a solar cell. 4
- c) What are the advantages of hybrid energy systems. 3
8. a) A 415V, 3 phase star connected induction motor draws a current of 22 Ampere at 0.8 power factor lagging under full load conditions. After connecting a star connected capacitor bank, the full load pf goes up to 0.9 lagging. Find the kVAr rating of the bank and capacitance of each capacitor. 7
- b) How energy audit leads to energy conservation? Give the classification of energy audit. 4
- c) List the measures taken to increase the operational efficiency of the pumps in agricultural sector. 3

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